

Multimodal Retrieval-Augmented Generation for Chart and Document Understanding: A Comprehensive Survey

Abstract

The convergence of Large Language Models (LLMs) and Vision-Language Models (VLMs) has catalyzed a paradigm shift in document intelligence, moving from static Optical Character Recognition (OCR) pipelines to dynamic, end-to-end Multimodal Retrieval-Augmented Generation (MRAG) systems. This survey provides a systematic analysis of MRAG methodologies specifically tailored for visually rich documents (VRDs), with a focus on the complex interplay between textual narratives, tabular data, and chart visualizations. We synthesize evidence from 300 recent academic works to construct a comprehensive taxonomy of the field, categorizing approaches by retrieval granularity, modality fusion strategies, reasoning architectures, and evaluation frameworks. Our analysis reveals a critical transition from text-centric retrieval, which often suffers from layout fragmentation and information loss, toward native visual retrieval mechanisms such as late-interaction multi-vector embeddings and layout-aware graph structures. We examine the emergence of agentic workflows that enable iterative, tool-augmented reasoning over multi-page and multi-document corpora, alongside novel training paradigms leveraging synthetic data and reinforcement learning to enhance factual grounding. Furthermore, we critically assess current benchmarks, highlighting persistent challenges in cross-modal alignment, hallucination mitigation, and the evaluation of complex reasoning tasks involving charts and tables. By integrating findings across retrieval mechanisms, generative strategies, and domain-specific applications, this survey outlines the current state-of-the-art and identifies pivotal open problems for future research in multimodal document understanding.

1. Introduction

The proliferation of digitally born and scanned documents in domains ranging from finance and healthcare to scientific research has created an unprecedented demand for automated systems capable of understanding, retrieving, and reasoning over heterogeneous content. Traditional document processing pipelines have historically relied on a modular architecture: extracting text via OCR, parsing layout heuristics, and feeding linearized text into language models. However, this approach fundamentally fails to capture the semantic richness embedded in the spatial arrangement of tables, the visual encoding of charts, and the cross-referential nature of multi-page documents [1], [2]. The advent of Multimodal Retrieval-Augmented Generation (MRAG) represents a transformative response to these limitations, integrating the generative capabilities of Large Multimodal Models (LMMs) with dynamic retrieval mechanisms to ground responses in precise visual and textual evidence [3], [4].

The evolution of MRAG can be characterized by three distinct phases. Early iterations (MRAG 1.0) treated multimodality as a preprocessing step, converting images and charts into text captions or structured JSON before indexing, a process prone to significant information loss and error propagation [5], [4]. The second phase (MRAG 2.0) introduced true multimodal embedding spaces, where queries and document patches are mapped to a shared vector space using contrastive learning, enabling direct image-to-image or text-to-image retrieval without intermediate transcription [6], [7]. The current frontier (MRAG 3.0) is defined by agentic and planning-based systems that decompose complex queries, iteratively retrieve evidence across modalities, and employ tool-use capabilities for fine-grained chart analysis and table reasoning [8], [9], [10].

A central challenge in this domain is the “modality gap” and the “granularity mismatch.” Queries

often target specific data points within a chart or a cell within a table, yet retrieval systems frequently operate at the page or chunk level, introducing noise that degrades generation quality [11], [12]. Furthermore, charts and tables require distinct reasoning capabilities: charts demand visual perception of trends and encodings, while tables necessitate structural understanding of rows, columns, and hierarchical relationships [13], [14]. Recent works have demonstrated that unified embedding models often struggle with these distinct requirements, prompting the development of specialized retrieval heads, hierarchical indexing, and graph-based knowledge representations [15], [16], [17].

This survey synthesizes insights from 300 relevant academic papers to provide a rigorous examination of MRAG for chart and document understanding. We organize the field into coherent methodological families, analyzing innovations in retrieval architectures, reasoning frameworks, and training strategies. Specifically, we explore: (1) **Retrieval Mechanisms**, contrasting late-interaction visual embeddings, layout-aware chunking, and graph-based indexing; (2) **Reasoning and Generation**, detailing agentic workflows, tool-augmented analysis, and citation-enforced generation; (3) **Data and Training**, covering synthetic data generation, instruction tuning, and reinforcement learning; and (4) **Evaluation**, critiquing current benchmarks and identifying gaps in assessing complex multimodal reasoning. By mapping the trajectory from simple retrieval-augmented pipelines to sophisticated, planning-driven agents, we aim to establish a foundational reference for researchers and practitioners navigating this rapidly evolving landscape.

2. Method

The methodological landscape of Multimodal RAG for document and chart understanding is diverse, spanning innovations in representation learning, retrieval architecture, reasoning strategies, and generative controls. We categorize these approaches into six primary dimensions: (1) Multimodal Representation and Embedding, (2) Granularity and Layout-Aware Retrieval, (3) Graph-Based and Structural Reasoning, (4) Agentic and Iterative Workflows, (5) Training Paradigms and Data Synthesis, and (6) Generation, Citation, and Faithfulness.

2.1 Multimodal Representation and Embedding Strategies

The foundation of any RAG system lies in its ability to represent queries and documents in a shared semantic space. In the multimodal context, this requires bridging the gap between textual queries and visual document content (pages, charts, tables).

2.1.1 Late-Interaction and Multi-Vector Embeddings A dominant paradigm in modern visual document retrieval is the shift from single-vector dense retrieval to multi-vector late-interaction mechanisms. Traditional dense retrievers compress a document page into a single vector, often losing fine-grained details crucial for chart and table analysis. In contrast, late-interaction models like **ColPali** encode document pages into a sequence of patch-level vectors, computing relevance via a MaxSim operation between query tokens and document patches [7]. This approach preserves local visual semantics, allowing the model to attend to specific chart axes or table cells without explicit OCR. Empirical evaluations on the ViDoRe benchmark demonstrate that ColPali significantly outperforms traditional captioning-based approaches and single-vector baselines [7], [6].

Building on this, **ColMate** introduces Masked OCR Language Modeling (MOLM) to optimize visual token representations, predicting masked OCR tokens from pooled visual embeddings to enhance semantic alignment [18]. Similarly, **Argus** incorporates a query-conditioned Mixture-of-

Experts (MoE) into the document encoder, dynamically routing visual patches based on query context to recover cross-encoder adaptivity while maintaining offline indexing efficiency [19]. These methods collectively suggest that preserving the spatial distribution of visual features is critical for high-fidelity retrieval in visually rich domains.

2.1.2 Unified and Co-Modality Embedding Spaces While late-interaction models excel at fine-grained matching, other approaches seek to unify text and image representations into a single embedding space to facilitate hybrid retrieval. **CMRAG** employs a Unified Encoding Model (UEM) with a Dual-Sigmoid Alignment loss to project queries, images, and parsed texts into a shared latent space, enabling complementary retrieval signals from both modalities [20]. **Unveil** utilizes knowledge distillation to transfer semantic richness from a visual-textual teacher (processing both OCR and images) to a visual-only student, achieving state-of-the-art performance while reducing inference latency [21].

However, the efficacy of unified spaces is debated. **SERVAL** argues that modern VLMs can generate sufficiently granular textual descriptions offline, allowing standard text encoders to bridge the modality gap without specialized multimodal training [22]. Conversely, **GME** demonstrates that instruction-tuned multimodal embedders, trained on a balanced mixture of single, cross, and fused-modal data, significantly outperform text-only or image-only specialists in universal retrieval tasks [23]. The choice between unified and specialized embeddings often depends on the specific domain constraints and the complexity of the visual content.

2.1.3 Sparse and Hybrid Retrieval Hybrid retrieval strategies combining sparse (lexical) and dense (semantic) signals have shown robustness in handling terminology mismatches and rare entities. **DocRetriever** extracts layout-aware sparse embeddings directly from VLM logit distributions, fusing them with dense scores to leverage both lexical precision and semantic understanding without relying on external OCR [24]. **LITTA** employs test-time query expansion, generating complementary query variants to address terminology gaps and aggregating results via Reciprocal Rank Fusion (RRF) [25]. Furthermore, **HEAVEN** proposes a two-stage hybrid approach where single-vector retrieval over visually summarized pages (VS-pages) serves as a coarse filter, followed by multi-vector reranking, achieving near-ColPali accuracy with drastically reduced computational cost [26].

2.2 Granularity and Layout-Aware Retrieval

Document understanding is inherently hierarchical. Retrieving entire pages for simple factual queries introduces noise, while retrieving individual tokens loses global context. Recent methods focus on adaptive granularity and layout awareness.

2.2.1 Block-Level and Region-Oriented Retrieval Shifting retrieval granularity from pages to semantic blocks has proven effective for precise query-content alignment. **LFRAG** segments pages into semantic blocks using DocLayout-YOLO and aggregates them via graph-based methods, employing a semantic-layout fusion encoder to integrate local block embeddings with global page context [11]. This approach achieves superior block-level recall compared to page-level baselines. Similarly, **RegionRAG** treats non-overlapping patches as fundamental retrieval units, utilizing hybrid supervision with gold-standard bounding boxes to isolate concise, context-complete visual segments [12]. **SCAN** advances this by performing coarse-grained semantic segmentation to detect

topically coherent regions, balancing context preservation with processing efficiency better than atomic detection [27].

2.2.2 Hierarchical and Structure-Aware Indexing Documents possess intrinsic hierarchical structures (e.g., section-page-element) that, when modeled explicitly, enhance retrieval coherence. **DMAP** constructs a hierarchical schema encoding these relationships, enabling tri-path retrieval that integrates structural hierarchy with dense textual and visual embeddings [28]. **HiKEY** reconstructs logical hierarchy trees via Document Hierarchical Parsing (DHP), using ancestry-aware subgraph assembly to preserve logical coherence during retrieval [29]. **MoDora** transforms OCR elements into a Component-Correlation Tree (CCTree), executing bottom-up cascade summarization to propagate hierarchical context [30]. These methods underscore that ignoring document structure leads to fragmented evidence retrieval, particularly in long documents.

2.2.3 Vision-Guided Chunking and Context Preservation Traditional chunking strategies often break semantic units across page boundaries. **Vision-Guided Chunking** processes PDFs in configurable page batches, maintaining cross-batch continuity via context injection containing previous batch headings and chunks [31]. **MultiDocFusion** employs a vision-based parser to reconstruct explicit section hierarchies, aligning visual segmentation with textual hierarchy to mitigate context loss [32]. **DocsRay** utilizes prompt-based pseudo-Table of Contents generation to define semantic boundaries, replacing fixed-size chunking with structure-aware segmentation [33]. These approaches ensure that retrieved contexts remain semantically coherent, which is vital for downstream reasoning.

2.3 Graph-Based and Structural Reasoning

For complex reasoning tasks involving multi-hop dependencies or cross-modal relationships, graph-based representations offer a structured alternative to flat vector retrieval.

2.3.1 Multimodal Knowledge Graphs (MMKG) Constructing knowledge graphs that integrate visual entities with textual concepts enables holistic reasoning. **MegaRAG** executes a two-round iterative graph construction, refining page-specific subgraphs to guide MLLMs in identifying missing cross-modal links [8]. **MMGraphRAG** transforms images into scene graphs and text into knowledge graphs, using spectral clustering for cross-modal entity linking to align visual and textual entities without manual annotation [34]. **DSRAG** separates conceptual frameworks from instance details in a hierarchical knowledge graph, enabling precise semantic pruning before vector search [35]. These systems demonstrate that structured representations reduce retrieval noise and facilitate complex inference chains.

2.3.2 Document Graphs and Topological Indexing Beyond semantic graphs, modeling the topological relationships between document pages enhances long-context understanding. **MoLoRAG** constructs a page graph where edges denote embedding similarity, enabling VLMs to traverse structural dependencies and identify logically critical evidence missed by vector similarity alone [36]. **MMRAG-DocQA** employs topological cross-page clustering to model long-distance dependencies, bridging modalities through co-occurrence and parent-page retrieval [37]. **G2-Reader** decouples evidence representation (Content Graph) from reasoning control (Planning Graph), allowing for structure-preserving retrieval and adaptive reasoning decomposition [38].

2.3.3 Chart-Specific Structural Representations Charts require specialized structural modeling to interpret visual encodings and data relationships. **ChartKG** represents charts as knowledge graphs comprising visual elements, property values, and data variables, enabling interpretable graph traversal for complex queries [39]. **MSG-Chart** constructs dual undirected graphs: a Visual Graph capturing spatial relations and a Textual Graph encoding semantic links between labels and axes, preserving object-wise integrity [40]. **StructChart** reformulates linear CSV data into Structured Triplet Representations (STR) to preserve positional relations critical for reasoning [41]. These methods highlight that treating charts as structured data rather than mere images significantly improves reasoning accuracy.

2.4 Agentic and Iterative Workflows

The complexity of document and chart understanding often exceeds the capacity of single-pass retrieval-generation pipelines. Agentic frameworks introduce planning, tool use, and iterative refinement.

2.4.1 Multi-Agent Collaboration Multi-agent systems distribute cognitive load across specialized roles. **MDocAgent** orchestrates five agents, including a “Critical Agent” that extracts key information to guide specialized modality agents, enabling granular attention to detail across text and visuals [42]. **VisDoMRAG** executes parallel textual and visual RAG pipelines, enforcing consistency constraints between separate reasoning chains to align modality-specific representations [43]. **ChartCitor** deploys six specialized agents for tasks ranging from chart-to-table extraction to entity captioning, decomposing attribution into structured retrieval followed by visual grounding [44]. **DocLens** utilizes a “Page Navigator” and “Element Localizer” to achieve high-recall page retrieval and fine-grained visual cropping [45].

2.4.2 Iterative Reasoning and Planning Iterative workflows allow models to refine queries and evidence dynamically. **ViDoRAG** integrates a Seeker-Inspector-Answer agent loop, using GMM-driven dynamic top-K selection to adapt retrieval depth per query [9]. **Doc-V*** formulates multi-page DocVQA as a sequential decision process, alternating between semantic retrieval and targeted page fetching actions based on a global thumbnail overview [46]. **CogPlanner** formalizes MRAG planning as a dual optimization of information acquisition and query reformulation, mimicking human cognitive processes to select retrieval modalities judiciously [10]. **AutothinkRAG** employs a Query Complexity Router to decouple perception and reasoning, routing simple queries to lightweight VLMs and complex ones to powerful LLMs [47].

2.4.3 Tool-Augmented Analysis Integrating external tools enhances precision in chart and table analysis. **ChartAgent** decomposes chart analysis into observable steps, orchestrating visual perception tools (detection, OCR) and reasoning tools (calculation) under a cost-gain trade-off policy [48]. **AgenticOCR** transforms OCR from static preprocessing to a dynamic, query-driven process, employing a unified tool primitive for region localization and content recognition [49]. **PDFTriage** utilizes function calling over document structural metadata, enabling an LLM triage agent to select precise structural elements (tables, figures) rather than plain text chunks [50]. These approaches demonstrate that tool augmentation mitigates the limitations of end-to-end generation in precise numerical and structural tasks.

2.5 Training Paradigms and Data Synthesis

The scarcity of high-quality, annotated multimodal document data has spurred innovations in synthetic data generation and efficient fine-tuning.

2.5.1 Synthetic Data Generation Synthetic data pipelines enable scalable training for diverse chart and document types. **ChartM3** employs a code-driven synthesis pipeline, generating executable Python scripts to create diverse chart-image-instruction triplets, preventing distributional convergence [51]. **ChartGen** utilizes a two-stage process where VLMs reconstruct seed charts into code, which is then iteratively augmented to diversify style and type while preserving semantics [52]. **SynthDoc** uses a hierarchical layout controller to programmatically render complex charts and tables, providing clean, structurally annotated training samples [53]. **Docs2Synth** implements an agent-based loop to generate and verify synthetic QA pairs targeting specific entities, bridging domain gaps without human labels [54].

2.5.2 Instruction Tuning and Adaptation Parameter-efficient fine-tuning (PEFT) and instruction tuning are critical for adapting base models to document tasks. **LayoutLLM** employs layout instruction tuning across hierarchical levels (document, region, segment) to enhance spatial comprehension [55]. **mPLUG-DocOwl** utilizes unified instruction tuning balancing language-only, general vision-language, and specific document instructions to enhance OCR-free comprehension [56]. **SV-RAG** employs a dual-adaptor architecture, utilizing MLLM hidden states as effective visual page embeddings optimized with contrastive loss [57]. **URaG** inserts a lightweight cross-modal retrieval module at early Transformer layers, leveraging the inherent coarse-to-fine reasoning of MLLMs [58].

2.5.3 Reinforcement Learning and Preference Optimization Reinforcement Learning (RL) is increasingly used to align generation with retrieval evidence and reduce hallucinations. **Look As You Think (LAT)** employs Group Relative Policy Optimization (GRPO) with rule-based rewards to enforce stepwise visual attribution in Chain-of-Thought reasoning [59]. **MM-Doc-R1** utilizes Similarity-based Policy Optimization (SPO) for dynamic baseline estimation in multi-turn RL, optimizing agent trajectories for long document QA [60]. **M2IO-R1** treats image insertion as a sequential decision problem, using GRPO to optimize global coherence in multimodal generation [61]. **Document Optimization via GRPO** treats document rewriting as a policy optimization problem, aligning content with retriever embedding spaces using counterfactual rewards [62].

2.6 Generation, Citation, and Faithfulness

Ensuring that generated answers are faithful to retrieved evidence and properly cited is paramount for trustworthy document understanding.

2.6.1 Visual Citation and Grounding Explicit citation mechanisms ground answers in specific document regions. **FinRAGBench-V** introduces fine-grained block-level visual citation via coordinate prediction, automatically evaluating citation accuracy [63]. **VISA** integrates bounding box prediction into VLMs, enabling precise visual evidence localization without OCR dependencies [64]. **Chain of Evidence (CoE)** generates structured JSON outputs containing answers and ordered evidence chains with pixel-level bounding box regression [65]. **ChartCitor** employs hierarchical entity captioning and a two-stage retrieve-then-rank mechanism to mitigate direct visual reasoning errors in chart QA [44].

2.6.2 Hallucination Mitigation and Verification Mitigating hallucinations requires rigorous verification loops. **FATHOMS-RAG** combines phrase-level recall metrics with nearest-neighbor embedding classifiers to distinguish factual statements from abstentions [66]. **Citation-Enforced RAG** implements hard-constraint citation enforcement and retrieval-confidence-driven abstention to reduce unsupported claims in fiscal documents [67]. **BayesRAG** reframes retrieval as Bayesian inference, using Dempster-Shafer theory to fuse evidence and penalize conflicting modalities [68]. **MRAG-Suite** introduces claim-level multimodal entailment judging to detect hallucinations across image and text modalities [69].

2.6.3 Interleaved and Structured Generation Generating interleaved text and visuals enhances explanatory power. **MRAMG** employs a bipartite graph matching algorithm for optimal sentence-image insertion positioning, demonstrating that LLM-based generation leveraging textual context outperforms direct MLLM input on simple datasets [70]. **Multimodal DeepResearcher** utilizes a Formal Description of Visualization (FDV) to convert charts into structured text for in-context learning, enabling high-fidelity report generation [71]. **EvidFuse** decouples visualization analysis from drafting, inserting visuals exactly when the narrative requires support via a real-time interactive multi-agent loop [72].

3. Challenges and Outlook

Despite significant advancements, Multimodal RAG for chart and document understanding faces several persistent challenges that hinder widespread deployment and robustness.

3.1 Evaluation Bottlenecks and Benchmark Limitations

Current benchmarks often fail to capture the complexity of real-world document reasoning. Many datasets, such as **DocVQA** and **ChartQA**, suffer from data contamination and saturation, leading to inflated performance metrics that do not generalize to harder evaluation protocols [3], [73]. Benchmarks like **MultiHaystack** reveal that retrieval performance degrades drastically (over 40 points) when shifting from unimodal to cross-modal settings due to embedding mismatches [74]. Furthermore, existing evaluations often lack diverse document types, focusing heavily on English-centric corpora and neglecting low-resource languages and right-to-left scripts, where performance drops significantly [75], [76]. The **ViDoRe V2** benchmark highlights significant generalization gaps in models overfitting to earlier distributions, necessitating blind contextual querying and cross-document tasks to truly assess robustness [77]. There is a critical need for benchmarks that evaluate multi-hop reasoning across charts and tables, such as **BRIDGE** and **MultiChartQA**, which expose severe degradation in parallel and sequential reasoning tasks [14], [78].

3.2 The OCR and Layout Fragility Problem

While end-to-end visual retrieval methods like ColPali reduce reliance on OCR, many systems still depend on preprocessing pipelines that are brittle under noise. **OCR Hinders RAG** demonstrates that semantic noise from OCR errors causes consistent performance degradation across all retrievers and LLMs, whereas formatting noise selectively impairs multimodal models [79]. **Lost in OCR Translation** finds that OCR-based pipelines with large LLMs generalize better to unseen degraded documents than fine-tuned VLMs, suggesting a trade-off between efficiency and robustness [80]. Layout analysis remains a bottleneck; errors in detecting table structures or chart boundaries propagate through the entire RAG pipeline, leading to hallucinated answers [81], [32]. Future

research must focus on robust, layout-aware parsing that can handle diverse and noisy document formats without catastrophic failure.

3.3 Complexity of Chart and Table Reasoning

Charts and tables present unique reasoning challenges that current MLLMs struggle to master. Models often excel at text extraction but fail significantly on chart interpretation, relying on surface patterns rather than visual reasoning [82], [83]. **EncQA** reveals that VLM performance varies significantly across visual encodings, with legend-based encodings (area, color) proving harder than position-based ones [84]. **FinDocMRE** identifies an “Analyst-Calculator Dichotomy,” where models exhibit strong semantic narrative synthesis but weak numerical precision and cross-page visual grounding [85]. Furthermore, **HighlightBench** shows that visual markups systematically bias model decisions, causing “Marker Hijacked” errors where predictions land on highlighted but logically irrelevant cells [86]. Addressing these issues requires specialized training objectives and architectural innovations that explicitly model numerical and structural relationships.

3.4 Scalability and Latency Constraints

Agentic and iterative workflows, while powerful, incur significant latency and computational costs. **ViDoRAG** notes that iterative multi-agent interactions increase latency compared to single-pass RAG [9]. **Failure is Feedback** reports API costs 9.64-16.60x higher and wall-clock time 5.20-6.71x slower for agentic traversal compared to non-agentic baselines [87]. **ColPali** inference latency (7.28s/query) hinders scalability for large document libraries [88]. Efficient retrieval mechanisms, such as **HEAVEN**’s hybrid approach or **ZipRerank**’s single-pass listwise scoring, are essential steps toward practical deployment, but further optimization is needed to support real-time applications on large-scale corpora [26], [89].

3.5 Safety, Hallucination, and Trustworthiness

Hallucination remains a critical risk, particularly in high-stakes domains like finance and healthcare. **ChartAttack** demonstrates the vulnerability of LLMs to malicious prompting in chart generation, where design-level misleaders cause significant accuracy drops [90]. **FATHOMS-RAG** and **MRAG-Suite** highlight the difficulty of distinguishing abstentions from hallucinations without model internals [69], [66]. While citation-enforced methods and verification loops improve faithfulness, they often come at the cost of completeness or latency [67], [63]. Future systems must integrate robust safety classifiers and uncertainty estimation mechanisms to ensure trustworthy outputs, especially when dealing with ambiguous or conflicting evidence.

3.6 Future Directions

Future research should prioritize the development of **universal multimodal embedders** that can handle diverse modalities and granularities without task-specific fine-tuning [23], [91]. There is a pressing need for **neuro-symbolic hybrid approaches** that combine the flexibility of neural retrieval with the precision of symbolic reasoning for tables and charts [92], [16]. Additionally, **cross-lingual and multilingual MRAG** systems must be advanced to support global document collections, addressing the current deficit in non-Latin script performance [93], [94]. Finally, the community must move towards **holistic evaluation frameworks** that assess not just answer accuracy but also retrieval precision, citation correctness, and reasoning transparency [69], [95].

4. Conclusion

Multimodal Retrieval-Augmented Generation has emerged as a pivotal framework for advancing document and chart understanding, transcending the limitations of traditional OCR-based pipelines. Through the synthesis of 300 academic works, this survey has illuminated the field’s rapid evolution from simple text-conversion strategies to sophisticated, agentic systems capable of iterative reasoning and fine-grained visual grounding. Key innovations in late-interaction embeddings, layout-aware retrieval, and graph-based reasoning have significantly enhanced the ability of models to navigate complex document structures and extract precise evidence from charts and tables.

However, the path to robust, scalable, and trustworthy MRAG systems is fraught with challenges. Persistent issues with OCR fragility, benchmark saturation, and the complexity of numerical reasoning in charts demand continued innovation. The integration of reinforcement learning, synthetic data generation, and rigorous citation mechanisms offers promising avenues for mitigating hallucinations and improving factual grounding. As the field matures, the focus must shift towards universal models that seamlessly integrate diverse modalities, efficient architectures that support real-time deployment, and comprehensive evaluation protocols that reflect the nuances of real-world document intelligence. By addressing these challenges, the research community can unlock the full potential of MRAG, enabling AI systems to truly understand and reason over the vast repository of human knowledge encoded in visually rich documents.

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